

Lesson 4 — Hypothesis Testing

From Classical Tests to Practical Decision Making

Applied Statistics Course

Lesson Roadmap

- 1 Foundations
- 2 z- and t-Tests for Means
- 3 Proportions and Chi-Squared Tests
- 4 Power and Sample Size (Optional)
- 5 Multiple Testing (Optional)
- 6 Nonparametric and Permutation Tests

Foundations

- 1 Foundations
- 2 z- and t-Tests for Means
- 3 Proportions and Chi-Squared Tests
- 4 Power and Sample Size (Optional)
- 5 Multiple Testing (Optional)
- 6 Nonparametric and Permutation Tests

Hypothesis Testing Foundations

Learning Objectives

- Translate scientific/business questions into H_0 (null hypothesis) vs. H_1 (alternative hypothesis)
- Describe Type I/II errors, significance level α (alpha), and power $1 - \beta$
- Interpret p-values correctly and connect to confidence intervals (**CI**)
- Recognize likelihood-ratio ideas underpinning classical tests

Prerequisites: Lessons 1–3 on sampling distributions, estimation, and CIs

Why Hypothesis Testing? A Concrete Scenario

The Drug Comparison Problem

You're testing whether Drug A reduces recovery time vs. Drug B.

Preliminary experiment (3 patients per drug):

- Drug A: 24, 28, 32 hours $\Rightarrow$ mean = 28 hours
- Drug B: 40, 45, 43 hours $\Rightarrow$ mean = 42.7 hours
- **Difference: 15 hours faster with Drug A!**

The Question

Should we conclude Drug A is better?

This concrete example will motivate every concept in hypothesis testing.

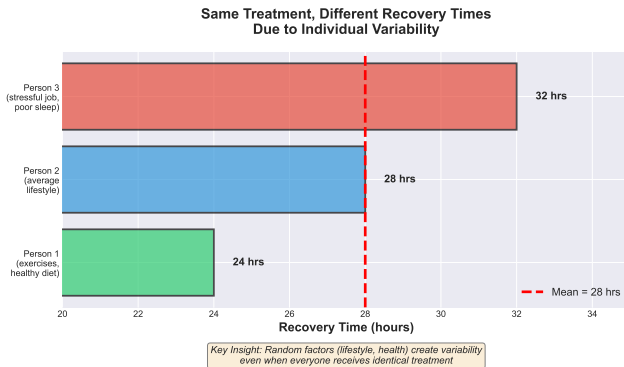
Why Results Vary: Individual Differences

The Problem

Not everyone recovers in the same time, even with identical treatment!

Confounding factors:

- Exercise level
- Diet quality
- Sleep patterns
- Stress levels
- Air pollution
- Baseline health
- Age & genetics



Random factors create variability $\Rightarrow$ hard to isolate drug's true effect

The Problem with Single Experiments

What if we repeat the experiment?

Trial 2 (new patients):

- Drug A: 45, 50, 48 hours $\Rightarrow$ mean = 47.7 hours
- Drug B: 30, 35, 33 hours $\Rightarrow$ mean = 32.7 hours
- **Now Drug B is 15 hours faster!**

Key Insight

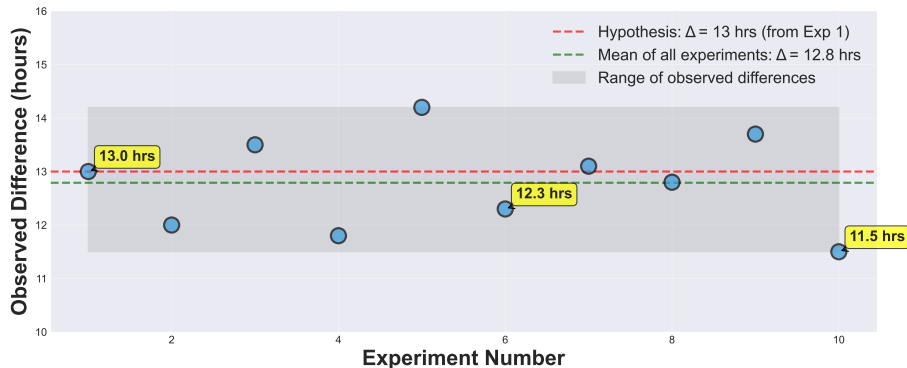
With small samples, random variability dominates.

One experiment is insufficient when dealing with random factors.

This is why we need statistical hypothesis testing.

The “Infinite Hypotheses” Problem

The Problem: Which Hypothesis Should We Test?
Observed differences vary across replicate experiments



□ Problem: Testing $\Delta = 13$ is arbitrary (why not 12? 13.5? 12.25?)
□ Solution: Test the NULL hypothesis ($\Delta = 0$)
→ Zero is the ONLY unambiguous value

The Elegant Solution: Test "No Difference"

Why Test the NULL Hypothesis ($\Delta = 0$)?

✗ THE PROBLEM

Infinitely many plausible hypotheses to test:

$\Delta = 12.0?$ $12.1?$ $12.25?$ $13?$

Which one is "correct"?



✓ THE SOLUTION

Test the NULL hypothesis:

$H_0: \Delta = 0$

(No difference between groups)

Why This Works:

- 1 Zero is the ONLY unambiguous value representing "no difference"
- 2 Doesn't require preliminary data to choose a hypothesis
- 3 Clear decision: either reject (difference exists) or fail to reject
- 4 Conservative approach - need strong evidence to claim an effect

Example: "Does Drug A differ from Drug B?"
Instead of testing "Drug A is 13 hours better" (arbitrary),
we test "There is NO difference" and see if data contradicts this.

Formalizing Hypotheses

Key Definitions

- **Null hypothesis (H_0):** The “no effect” or “status quo” claim we test against
- **Alternative hypothesis (H_1):** What we want to detect; the claim of an effect or difference

Common Examples

- **Means:** $H_0 : \mu_A = \mu_B$ vs. $H_1 : \mu_A \neq \mu_B$
- **One-sided:** $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$
- **Proportions:** $H_0 : p = p_0$ vs. $H_1 : p \neq p_0$
- **Independence:** H_0 : variables are independent

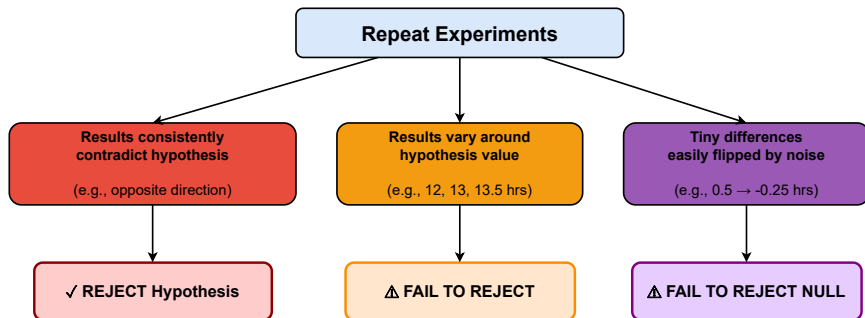
Drug Example

$H_0: \mu_A - \mu_B = 0$ (no difference in recovery time)

$H_1: \mu_A - \mu_B \neq 0$ (there is a difference)

Making Decisions: Reject vs. Fail to Reject

Hypothesis Testing Decision Framework



Key Points:

- We REJECT when data strongly contradicts our hypothesis
- We FAIL TO REJECT when data is consistent with hypothesis (but not proof!)
- We NEVER say "accept" - absence of evidence $\neq$ evidence of absence

Test Statistic and Sampling Distribution

The Key Ingredient

Choose a **test statistic** $T(X)$ — a summary value computed from data that measures evidence against H_0 .

Drug Example (Two-Sample t-test)

$$T = \frac{\bar{X}_A - \bar{X}_B}{S_p \sqrt{1/n_A + 1/n_B}}$$

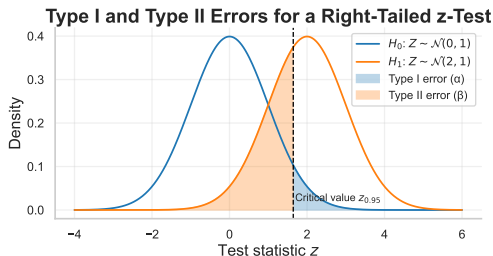
where S_p is the pooled standard deviation.

Under H_0 : $T \sim t_{n_A+n_B-2}$ (if data are normally distributed)

How It Works

- Large $|T| \Rightarrow$ observed difference is many standard errors from 0
- Compare to critical value or compute p-value
- Reject if evidence is “extreme enough”

Two Types of Errors



Type I Error (False Positive)

Reject H_0 when H_0 is true

Probability = α (typically 0.05)

Type II Error (False Negative)

Fail to reject H_0 when H_1 is true

Probability = β

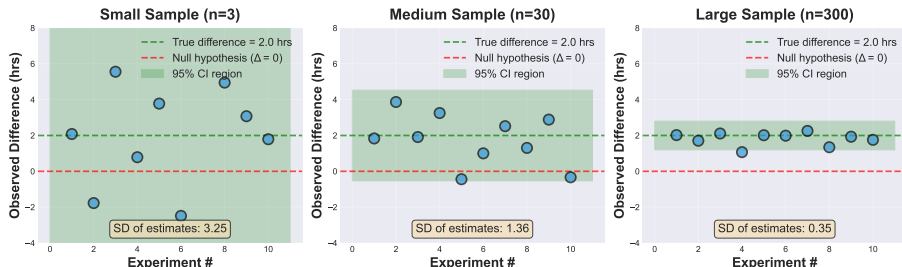
The Critical Role of Sample Size

Statistical Power

Power = $1 - \beta$ = Probability of correctly detecting a real effect when it exists.

Target: typically 0.80 (80% chance to detect real effect)

Sample Size Effect: Larger Samples → More Stable Estimates → Stronger Evidence



Key Insight: With small samples, random factors dominate. With large samples, true effects become clear.

Key Insight

Larger samples $\Rightarrow$ more stable estimates $\Rightarrow$ higher power

Planning for Adequate Power

Before Collecting Data

- 1 Specify the **minimal detectable effect** you care about
- 2 Choose significance level α (typically 0.05)
- 3 Target power level (typically 0.80)
- 4 Calculate required sample size using power formulas

Drug Study Example

Want to detect 10-hour difference with 80% power at $\alpha = 0.05$

Assume $\sigma \approx 15$ hours

$\Rightarrow$ Need $n \approx 36$ patients per group

Takeaway

Power analysis is part of test design: it balances sensitivity, cost, and time *before* any data is collected.

Summary of the Testing Setup

Element	Description	Variables / Expressions
Question	Compare the redesigned onboarding (B) to the current flow (A).	Variant A / Variant B
Data	Completion treated as a Bernoulli outcome for each sign-up; summarize with sample proportions.	$X_A, X_B \in \{0, 1\}$ $\hat{p}_A = X_A/n_A, \hat{p}_B = X_B/n_B$
Parameters	True completion rates for each variant.	p_A, p_B
Hypotheses	One-sided test focused on improvement.	$H_0 : p_B - p_A \leq 0$ $H_1 : p_B - p_A > 0$
Test statistic	Standardized difference in proportions using pooled variance.	$Z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1 - \hat{p})(1/n_A + 1/n_B)}}$ $\hat{p} = (X_A + X_B)/(n_A + n_B)$
Reference	Large-sample Normal approximation from Lesson 3.	$Z \approx \mathcal{N}(0, 1)$
Decision rule	Reject H_0 (ship B) when Z exceeds the one-sided critical value at level α .	Threshold $z_{1-\alpha}$ / p-value $< \alpha$
Risks	Type I: ship a worse flow Type II: miss a true improvement	–

From Question to Test

Workflow

- 1 Formulate H_0 (status quo) and H_1 (change of interest)
- 2 Choose a test statistic $T(X)$ and its reference distribution under H_0
- 3 Compute observed statistic t_{obs} from data
- 4 Quantify evidence with a p-value or rejection region
- 5 Report decision together with effect size and confidence interval

Diagnostic Questions

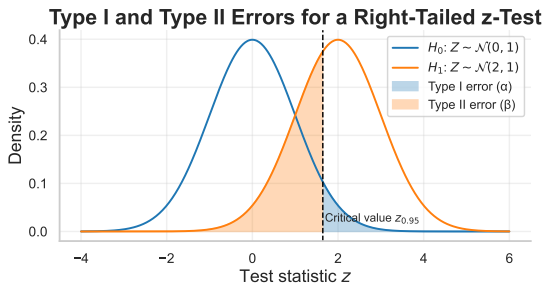
- What outcomes correspond to costly false positives/negatives?
- Are distributional assumptions (Normal, Binomial, independence) defensible?
- How will results influence operational or scientific decisions?

Type I and Type II Errors

Key Definitions

- **Type I:** reject H_0 when true $\Rightarrow$ rate = α
- **Type II:** fail to reject H_0 when H_1 holds $\Rightarrow$ rate = β
- **Power:** $1 - \beta$, detect effect when present

Example: Type I = launch worse flow; Type II = keep inferior status quo.



Understanding p-Values

Definition

The **p-value** is the probability, **assuming H_0 is true**, of observing data as extreme or more extreme than what we actually observed.

Example: Testing a Coin

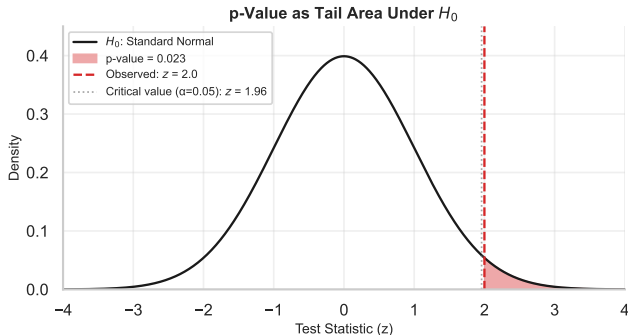
H_0 : fair ($p = 0.5$) vs. H_1 : biased. Flip 10 times, get 9 heads.

p-value = 0.021: If fair, ≥ 9 heads happens only 2.1% of the time.
Since $0.021 < 0.05$, reject H_0 (likely biased).

Interpretation

- **Small p-value ($p < 0.05$)**: Data would be very unlikely under H_0
 $\Rightarrow$ Evidence against H_0 , so we reject it
- **Large p-value ($p > 0.05$)**: Data is consistent with H_0
 $\Rightarrow$ Not evidence for H_0 , just absence of evidence against it

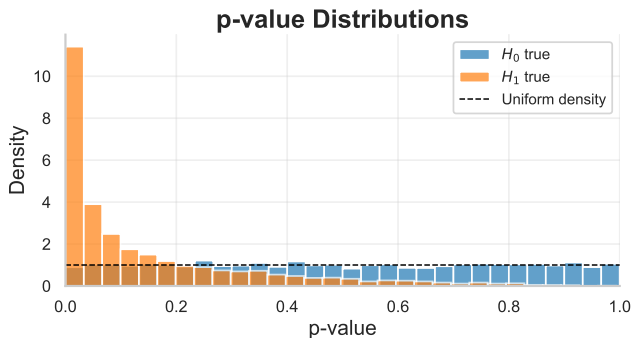
p-Value Visualization



The **shaded red area** represents the p-value: the probability of observing a test statistic this extreme (or more extreme) **if H_0 were true**.

Key insight: The smaller this area, the more “surprising” our data would be under H_0 , and thus the stronger the evidence against H_0 . When this area is less than α (e.g., 0.05), we reject H_0 .

p-Value Distributions Across Many Tests



Blue (H_0 true): p-values are uniformly distributed. About 5% fall below 0.05 (Type I error rate).

Orange (H_1 true): p-values cluster near 0. Most are <0.05 , showing high power to detect the effect.

Key insight: Under H_0 , all p-values are equally likely. Under H_1 , small p-values dominate.

What p-Values Are NOT

Common Misconceptions

- ❶ **NOT** the probability that H_0 is true
(p-values assume H_0 is true; they don't test whether it's true)
- ❷ **NOT** the probability results are “due to chance”
(informal and ambiguous phrasing)
- ❸ **NOT** a measure of effect size or importance
(small p-value $\neq$ large or meaningful effect)
- ❹ **NOT** proof of anything
(statistical tests provide evidence, not certainty)

Best Practice

Always report: p-value + effect size + confidence interval + context

Tests and Confidence Intervals

Duality Principle

For many one-parameter problems, *reject* $H_0 : \theta = \theta_0$ at level α **iff** θ_0 lies **outside** the $(1 - \alpha)$ confidence interval for θ .

Connection to p-Values

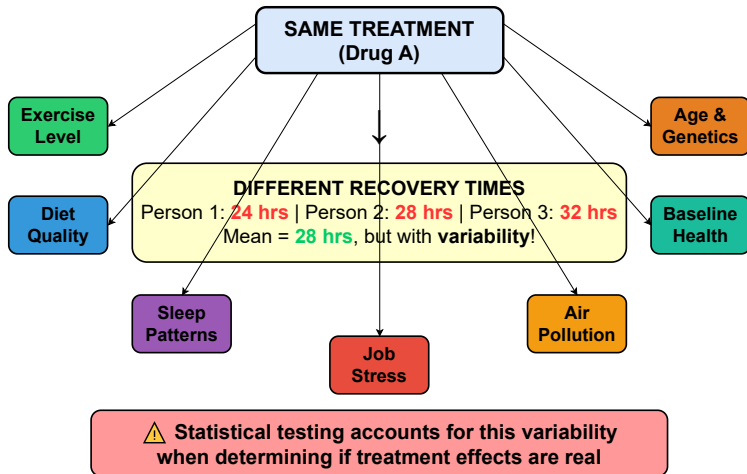
- **p-value** $< \alpha \Leftrightarrow \theta_0$ outside $(1 - \alpha)$ CI $\Rightarrow$ reject H_0
- **p-value** $> \alpha \Leftrightarrow \theta_0$ inside $(1 - \alpha)$ CI $\Rightarrow$ don't reject H_0
- **p-value** $= \alpha \Leftrightarrow \theta_0$ at boundary of $(1 - \alpha)$ CI

Why This Matters

- CIs are more informative: show decision *and* effect size
- Same α gives same decision via test or CI
- In labs, verify test decisions match CI inclusion of null value

Understanding Variability in Experiments

Why Do Recovery Times Vary? Uncontrolled Confounding Factors



Checklist Before Running a Test

- Clarify the consequence of each error type with stakeholders
- Verify model assumptions (independence, distribution family)
- Pre-plan significance level and desired power (Lesson 4 power module)
- Pair test results with effect sizes and confidence intervals
- Consider confounding factors and sources of variability

Preview of Hands-On Lab

Simulate Type I/II rates, plot p-value histograms under H_0/H_1 , and relate empirical findings to theory.

Summary

Key Takeaways

- Hypothesis testing formalizes evidence comparison between H_0 and H_1
- Type I/II errors quantify risk; power connects to sample-size planning
- p-values measure surprise under H_0 and must be contextualized with intervals
- Likelihood ratios motivate many classical procedures studied next

For Deeper Understanding

See `material.md` Section 1 for detailed explanations, examples, and the connection to confidence intervals.

- Casella & Berger (2002), *Statistical Inference* — Chapters 8–9
- Lehmann & Romano (2005), *Testing Statistical Hypotheses*

z- and t-Tests for Means

- 1 Foundations
- 2 z- and t-Tests for Means
- 3 Proportions and Chi-Squared Tests
- 4 Power and Sample Size (Optional)
- 5 Multiple Testing (Optional)
- 6 Nonparametric and Permutation Tests

Learning Objectives

- Distinguish when to use **z-tests** (known variance) vs. **t-tests** (unknown variance)
- Interpret test statistics, p-values, and confidence intervals jointly
- Understand pooled vs. **Welch's t-test** (default for unequal variances) and paired designs
- Report practical significance using effect sizes (e.g., **Cohen's d**)

*Datasets: heights/weights sample
(shared/data/heights_weights_sample.csv)*

One-Sample Location Tests

z-Test (variance known)

z-test: Test for population mean when σ is known.

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \quad (H_0).$$

Reject $H_0 : \mu = \mu_0$ if $|Z| > z_{1-\alpha/2}$.

t-Test (variance unknown)

t-test: Test for mean when σ is unknown (most common).

$$T = \frac{\bar{X}_n - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad (\text{Normal data}).$$

df (degrees of freedom): $n - 1$ = number of independent values.

Converges to z-test as n grows.

Two Independent Samples

Welch's t-Test (default)

Welch's t-test: Compare means without equal variance assumption.

$$T = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{S_A^2/n_A + S_B^2/n_B}}, \quad \text{df via Welch-Satterthwaite.}$$

Handles different variances/sizes. **Recommended default.**

Pooled t-Test (equal variances)

Pooled t-test: Uses combined variance.

$$T = \frac{\bar{X}_A - \bar{X}_B}{S_p \sqrt{1/n_A + 1/n_B}}, \quad S_p^2 = \frac{(n_A - 1)S_A^2 + (n_B - 1)S_B^2}{n_A + n_B - 2}.$$

More powerful when equal-variance holds. Diagnose via plots.

Paired Designs

Paired t-Test

Paired t-test: Compare paired/matched observations (e.g., before/after).

- One-sample test on differences $D_i = X_{Ai} - X_{Bi}$
- Controls for individual differences (each person is their own control)
- More powerful: removes baseline variability

Example: Blood Pressure Treatment

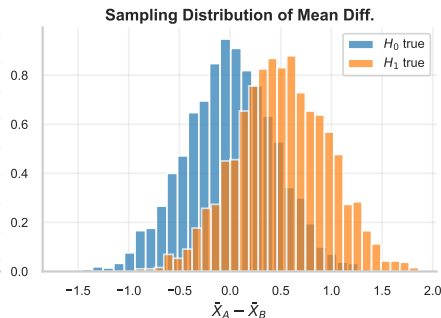
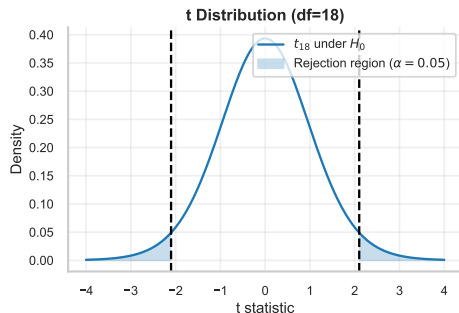
Patient	Before	After	Difference
A	140	130	-10
B	135	128	-7
C	150	142	-8

Test: Is mean difference $\bar{D} = 0$? (paired t-test on differences)

When to Use

Repeated measurements (before/after), matched pairs (twins), or any design with natural pairing

Visualizing Test Statistics



Left: t_{18} sampling distribution with rejection regions. Right: Monte Carlo comparison showing how the mean difference distribution shifts under H_1 .

Effect Sizes and Reporting

Cohen's d (Standardized Effect Size)

$$d = \frac{\bar{X}_A - \bar{X}_B}{S_p} = \frac{\text{signal (observed difference)}}{\text{noise (background variability)}}$$

Intuition: Signal-to-noise ratio how large is the difference compared to the randomness in your data?

Interpretation: Small ≈ 0.2 , Medium ≈ 0.5 , Large ≈ 0.8 (Cohen's conventions)

Example Report

"Drug A reduced recovery time by 4.7 hours (95% CI: [2.1, 7.3], $t = 3.45$, $p = 0.001$, $d = 0.67$). This represents a medium, clinically meaningful effect."

Reporting Checklist

Point estimate + CI + p-value + effect size + practical interpretation + assumption checks

Common Pitfalls

Watchlist

- Using pooled t-test without verifying equal variances
- Ignoring dependence structure (treating paired data as independent)
- Reporting only p-values without intervals or effect sizes
- Over-interpreting statistically significant but tiny effects (large n)

Lab Connection

Lesson notebook explores Welch vs. pooled tests, paired designs, and visual diagnostics on the heights dataset.

Summary

Key Takeaways

- z-tests require known variance (rare); t-tests cover most practical cases
- Welch t-test is the safe default for independent samples
- Paired analysis boosts power when measurements are naturally linked
- Always combine statistical significance with effect sizes and diagnostics

For Deeper Understanding

See `material.md` Section 3 for Python examples, detailed formulas, and effect size calculations.

- Student (1908), “The probable error of a mean”
- Sawilowsky (2009), “New effect size rules of thumb”

Proportions and Chi-Squared Tests

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Binary Outcomes and Contingency Tables

Learning Objectives

- Conduct one- and two-sample **proportion tests** (**A/B testing**)
- Apply **chi-squared** goodness-of-fit (**GOF**) and independence tests
- Diagnose when to use exact methods (**Fisher's exact test**) for sparse data
- Interpret **lift**, residuals, and effect sizes for categorical data

*Datasets: `shared/data/ab_test_clicks.csv`,
`shared/data/manufacturing_defects.csv`*

One-Proportion z-Test

Setup

Proportion: Fraction of successes in binary outcomes (e.g., conversion rate).

For $X \sim \text{Bin}(n, p)$, test $H_0 : p = p_0$ using

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \approx \mathcal{N}(0, 1) \quad (np_0, n(1 - p_0) \gtrsim 10).$$

where $\hat{p}$ is the sample proportion.

Practical Notes

- Use Wilson or Agresti–Coull intervals for better coverage with small n
- For very small counts, prefer an exact Binomial test
- Interpret results in terms of rate change or percentage point lift

Two-Proportion (A/B) Test

A/B Testing

A/B testing: Compare two variants to determine which performs better.

Pooled z-Test

$$Z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1 - \hat{p})(1/n_A + 1/n_B)}}, \quad \hat{p} = \frac{X_A + X_B}{n_A + n_B}.$$

Reject $H_0 : p_A = p_B$ when $|Z| > z_{1-\alpha/2}$.

Implementation Tips

- Report CI for $p_B - p_A$ (Wilson score)
- **Lift:** $(p_B - p_A)/p_A \times 100\%$
- Use Fisher's exact when counts < 5

Chi-Squared Procedures

The Chi-Squared Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (\text{sum of standardized squared deviations})$$

Intuition: Measures total discrepancy between observed and expected. Large $\chi^2 \Rightarrow$ reject H_0 .

Two Main Uses

- 1. Goodness-of-Fit:** Does ONE variable match expected distribution? (e.g., are dice rolls uniform?)
- 2. Independence:** Are TWO variables independent? (e.g., is gender independent of product choice?)

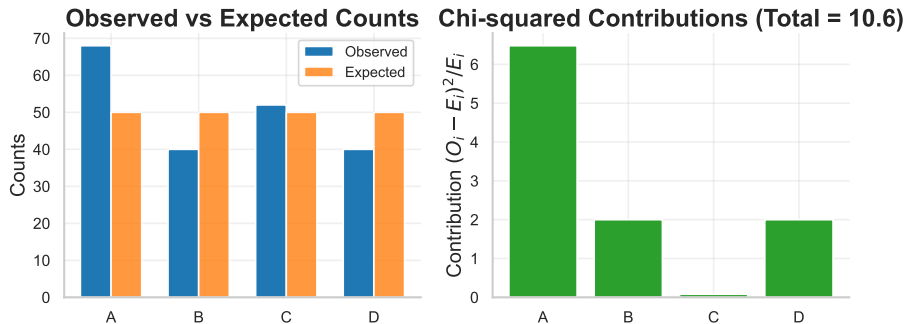
Example: 2x2 Independence Test

	Clicked	Didn't	Total
Ver. A	45	55	100
Ver. B	60	40	100

Test: Is click rate independent of version? $df =$

1

Chi-Squared Contributions — Visual



Left: observed vs. expected counts; Right: cell-wise contributions $(O_i - E_i)^2 / E_i$ identifying categories out of line with expectation.

Fisher's Exact Test

What It Is

Fisher's exact test: Exact test for 2×2 tables. Computes exact probability (no large-sample approximation). Use when chi-squared assumptions fail.

Example: Small Sample (Expected Count < 5)

	Success	Failure	Total
Treatment	8	2	10
Control	3	7	10

$(10 \times 9)/20 = 4.5 < 5 \Rightarrow$ Use Fisher's

Expected: $(10 \times 11)/20 = 5.5$,

Odds Ratio (OR)

Measures strength of association: $OR = 1$ (no effect), $OR > 1$ (positive association), $OR < 1$ (negative).

Example: $OR = 2.5$ means treatment increases odds of success by 150%.

Reporting Categorical Results

Checklist

- Provide sample sizes, counts, conversion/defect rates
- Include CI for rate difference or ratio (Wilson/Agresti–Coull)
- Visualize observed vs. expected counts for stakeholders
- Discuss both statistical and practical significance (lift, operational impact)

Lab Connection

Notebook replicates CTR analysis, Wilson intervals, chi-squared residual heatmaps, and Fisher's exact test for sparse tables.

Summary

Key Takeaways

- Proportion z-tests extend z/t logic to binary outcomes (with pooled variance under H_0)
- Chi-squared tests generalize to multi-category goodness-of-fit and independence problems
- Fisher's exact test safeguards inference when sample sizes are small
- Always complement significance tests with effect sizes (lift, odds ratios) and residual diagnostics

For Deeper Understanding

See `material.md` Section 4 for A/B testing examples, Python code, and detailed chi-squared applications.

- Agresti (2019), *An Introduction to Categorical Data Analysis*
- Brown, Cai, & DasGupta (2001), "Interval estimation for a binomial proportion"

Power and Sample Size (Optional)

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Why Power Matters

Optional Topic

This section is optional and may be skipped depending on course scope. See `material.md` Appendix A for comprehensive coverage of power analysis.

Learning Objectives

- Define statistical power and understand the α/β trade-off
- Quantify how effect size, variability, and sample size interact
- Use closed-form approximations and software (statsmodels) for planning
- Communicate minimal detectable effect (MDE) to stakeholders

Minimum Detectable Effect (MDE)

MDE: The smallest effect size that matters *practically*, not just statistically.

Example: “We care if the drug reduces recovery by ≥ 8 hours (shorter reductions don’t justify the cost).”

Power Function Basics

Definition

Statistical power = $\mathbb{P}(\text{reject } H_0 \mid H_1 \text{ true}) = 1 - \beta$.

Probability of correctly detecting a real effect when it exists.

Drivers

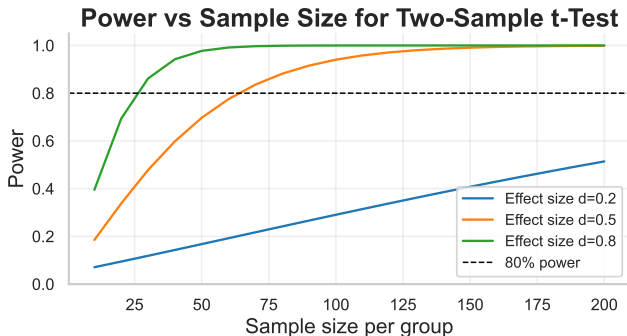
- **Effect size:** bigger differences easier to detect
- **Sample size:** more data reduces sampling variability
- **Variability:** higher noise lowers power
- **Significance level α :** stricter α reduces power

Plan experiments with target α (e.g., 0.05) and power (e.g., 0.8) given practically meaningful effect.

Power Curves

What Are Power Curves?

Power curve: Graph showing how statistical power changes with sample size or effect size.



Power vs. sample size for standardized effects $d \in \{0.2, 0.5, 0.8\}$. Note diminishing returns: beyond a point, doubling sample adds little additional power.

Back-of-the-Envelope Formulas

Two-Sample Means (balanced)

Given standardized effect $\delta = (\mu_B - \mu_A)/\sigma$,

$$n \approx 2 \left(\frac{z_{1-\alpha/2} + z_{\text{power}}}{\delta} \right)^2.$$

Two-Sample Proportions

$$n \approx 2 \frac{\left(z_{1-\alpha/2} \sqrt{2\bar{p}(1-\bar{p})} + z_{\text{power}} \sqrt{p_A(1-p_A) + p_B(1-p_B)} \right)^2}{(p_B - p_A)^2}$$
$$\bar{p} = \frac{p_A + p_B}{2}$$

Use analytical formulas for quick estimates, then refine with software.

statsmodels Power Solvers

- `TTestIndPower`: means (balanced or ratio $\neq 1$)
- `NormalIndPower`: proportions (z approximations)
- Output sample size, power, or effect size given the other two

Workflow

- 1 Specify α , desired power, target effect/**MDE**
- 2 Solve for sample size or MDE for fixed n
- 3 Summarize plainly (e.g., “detect 3 pp lift”)

MDE: Smallest effect reliably detectable with given power.

Communicating Power Analyses

Key Talking Points

- “With $n = 400$ per variant we can detect a 3 pp lift with 80% power”
- “To detect a 1 pp lift we would need roughly 9x more traffic”
- Highlight trade-offs: detecting smaller effects requires disproportionately more data
- Provide visual aids (power curves, MDE vs. sample size)

Lab Connection

Notebook replicates power curves, MDE calculations, and Monte Carlo validation for both means and proportions.

Summary

Key Takeaways

- Power analysis aligns statistical design with practical goals
- Effect size, sample size, variability, and α jointly determine power
- Use analytical approximations for intuition, but confirm with software
- Communicate results in terms stakeholders understand (MDE, required traffic)

For Comprehensive Coverage

See `material.md` Appendix A for detailed formulas, power curves, and advanced topics in power analysis.

- Cohen (1988), *Statistical Power Analysis for the Behavioral Sciences*
- Lipsey & Hurley (2009), “Design sensitivity: Statistical power for applied experimental research”

Multiple Testing (Optional)

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Multiplicity in Modern Experiments

Optional Topic

This section is optional and may be skipped depending on course scope. See `material.md` Appendix B for detailed coverage of multiple testing corrections.

Learning Objectives

- Understand why testing many hypotheses inflates false positives
- Distinguish between family-wise error rate (FWER) and false discovery rate (FDR)
- Apply Bonferroni, Holm, and Benjamini–Hochberg adjustments
- Communicate adjustment choices to product or research stakeholders

Scenario: A/B platform monitoring dozens of metrics per experiment

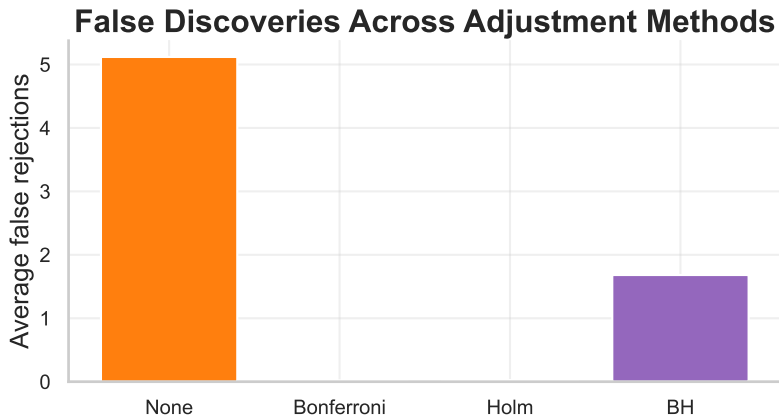
Why Adjust?

Inflation of False Positives

With m independent nulls at level α , expected false positives = $m\alpha$.

Risk Perspective

- **FWER:** Probability of ≥ 1 false discovery
- **FDR:** Expected proportion of false discoveries
- **FWER control:** Conservative, no false positives
- **FDR control:** Less conservative, bounds proportion
- Choose by decision cost: regulatory vs exploratory



Simulation (50% nulls). No adjustment inflates false positives; Holm and Bonferroni curb FWER; BH controls FDR while retaining more true signals.

Bonferroni Correction

Bonferroni: Test each hypothesis at level α/m (where m = number of tests).

Reject H_i if $p_i \leq \alpha/m$. Simple, valid under dependence, but conservative.

Holm Step-Down

Holm procedure: Less conservative step-down method.

Order p-values $p_{(1)} \leq \dots \leq p_{(m)}$; find first index k such that $p_{(k)} > \alpha/(m - k + 1)$ and accept thereafter.

Use for confirmatory analyses where any false positive is costly.

Benjamini–Hochberg (BH) Procedure

BH procedure: Controls FDR while allowing more discoveries than FWER methods.

- 1 Sort p-values $p_{(1)} \leq \dots \leq p_{(m)}$
- 2 Find largest k with $p_{(k)} \leq k\alpha/m$
- 3 Reject $H_{(1)}, \dots, H_{(k)}$

Practical Notes

- Controls expected proportion of false discoveries
- Suited for large-scale screening (omics, product telemetry)
- **q-value:** FDR-adjusted p-value (analogous to adjusted p-value for FDR)
- Visualize sorted p-values against BH threshold line for transparency

Communicating Results

Checklist

- State number of hypotheses tested and adjustment method used
- Report both raw and adjusted p-values (or **q-values** for FDR)
- Prioritize effect sizes and practical impact among surviving metrics
- Document rationale for chosen error rate (FWER vs FDR)

Lab Connection

Notebook runs null-only and mixed-signal simulations, compares adjustments, and plots q-value thresholds.

Summary

Key Takeaways

- Multiplicity inflates false positive risk; adjustment aligns error rates with tolerance
- Bonferroni/Holm control FWER; BH controls FDR with higher discovery power
- Always disclose number of tests and correction method in reports
- Simulation is a persuasive tool for explaining multiplicity to stakeholders

For Comprehensive Coverage

See `material.md` Appendix B for detailed explanations, when to use each method, and Python examples.

- Holm (1979), “A simple sequentially rejective multiple test procedure”
- Benjamini & Hochberg (1995), “Controlling the false discovery rate”

Nonparametric and Permutation Tests

- 1 Foundations
- 2 z- and t-Tests for Means
- 3 Proportions and Chi-Squared Tests
- 4 Power and Sample Size (Optional)
- 5 Multiple Testing (Optional)
- 6 Nonparametric and Permutation Tests

Learning Objectives

- Identify when parametric assumptions (Normality, equal variance) break down
- Apply **rank-based tests** (**Mann–Whitney U**, **Wilcoxon signed-rank**)
- Construct **permutation tests** for custom statistics
- Compare conclusions across parametric and **nonparametric** approaches

Datasets: heights/weights sample; simulated skewed distributions

When to Go Nonparametric

What Are Nonparametric Tests?

Nonparametric tests: Tests that do not assume a specific distribution (e.g., normal distribution).

More robust to violations of assumptions.

Indicators

- Heavy-tailed or skewed data (income, response times)
- Outliers that violate Normality assumptions
- Ordinal or rank-based measurements
- Small sample sizes where CLT support is weak

Nonparametric tests trade some efficiency under ideal Normal conditions for robustness elsewhere.

Rank-Based Tests

What Are Ranks?

Ranking: Convert values to their position when sorted (smallest = 1, largest = n).

Why? Robust to outliers and works with ordinal data.

Example: Mann–Whitney U (Independent Groups)

Group A: [35, 42, 38] Ranks: [1, 3, 2] (sum = 6)

Group B: [50, 55, 48] Ranks: [5, 6, 4] (sum = 15)

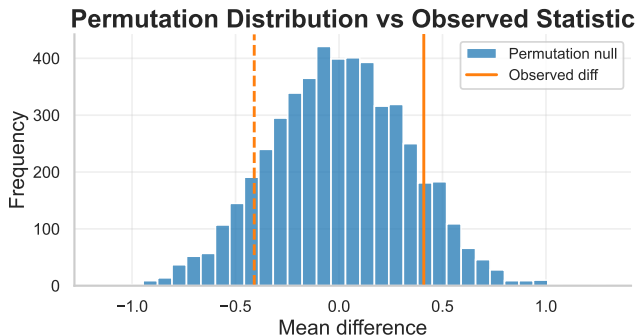
Test: Do groups have same distribution? (B ranks consistently higher)

Wilcoxon Signed-Rank (Paired Data)

Alternative to paired t-test. Rank absolute differences, then compare signed rank sums.

Effect size: $r_{rb} = 1 - 2U/(n_{ANB})$ ranges from -1 to 1 (like correlation).

Permutation Tests



Blue histogram: Null distribution from 5000 shuffles of group labels.

Solid orange: Observed mean difference from actual data.

Dashed orange: Negative observed (for two-tailed test).

p-value: Proportion of permutations as extreme as observed. No Normality assumption needed.

Constructing a Permutation Test

Key Idea

Under H_0 (no difference), group labels are arbitrary, so we can shuffle them.

Example: Two-Sample Test

Group A: [5, 7, 6], Group B: [8, 9, 10] Observed diff = 3.0

Shuffle 1: A=[5,9,6], B=[7,8,10] diff = 1.0

Shuffle 2: A=[8,7,10], B=[5,9,6] diff = -1.0

... repeat 5000 times ...

p-value = (count with $|\text{diff}| \geq 3.0$) / 5000

Algorithm

1. Compute observed statistic
2. Shuffle labels, recompute (5000+ times)
3. p-value = fraction as extreme as observed
4. Works for any statistic

Comparing Approaches

Recommended Workflow

- Start with parametric test (t-test) and check assumptions
- Run nonparametric/permutation counterpart for robustness
- If conclusions diverge, investigate data features (skew, outliers)
- Report both results with justification for chosen decision rule

Lab Connection

Notebook contrasts Welch t-test, Mann–Whitney U, and permutation test on the heights dataset + simulated skewed data.

Summary

Key Takeaways

- Nonparametric tests provide robustness when distributional assumptions fail
- Rank-based methods analyze relative ordering, targeting medians/effects on distribution
- Permutation tests offer exact finite-sample control under exchangeability
- Reporting both parametric and robust results builds confidence in findings

For Deeper Understanding

See `material.md` Section 5 for when to use nonparametric tests, Python examples, and detailed comparisons.

- Hollander, Wolfe, & Chicken (2014), *Nonparametric Statistical Methods*
- Good (2005), *Permutation, Parametric and Bootstrap Tests of Hypotheses*

For Deeper Understanding

Course Materials

- **Slides:** Overview and key concepts
- **material.md:** Detailed explanations and theory
- **TEST_SELECTION_GUIDE.md:** Decision flowchart

Where to Find More

- [Section 1](#) Foundations (p-value interpretation, error types)
- [Section 2](#) Test Selection Guide (flowchart, common tests)
- [Section 3](#) Tests for Means (Python examples)
- [Section 4](#) Proportions & Categorical (A/B testing)
- [Section 5](#) Non-Parametric Tests (robust alternatives)
- [Appendix A](#) Power Analysis (optional)
- [Appendix B](#) Multiple Testing (optional)

Glossary: Key Terms & Abbreviations

H_0 Null hypothesis — status quo, “no effect”

H_1 Alternative hypothesis — what we want to detect

α Significance level, Type I error rate (typically 0.05)

β Type II error rate (false negative)

Power $1 - \beta$, probability of correctly rejecting H_0 when H_1 is true

p-value Probability of data as extreme (or more) under H_0

CI Confidence Interval

df Degrees of freedom

Cohen's d Standardized effect size (0.2/0.5/0.8 = small/med/large)

GOF Goodness-of-Fit test

FWER Family-Wise Error Rate

FDR False Discovery Rate

MDE Minimum Detectable Effect